Department of Economics

MSc Extended Essay Cover Sheet

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TITLE	**The Effects of Political Hierarchy Change on the Local Economic Performance**

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Abstract

In this paper, we investigate the impact of centralization on local economic performance by analyzing the political hierarchy reform in China that transformed counties into districts between 2000 and 2013. In this county-upgrade centralization process, some counties lost their political autonomy with respect to the fiscal, administrative, and personnel decision-making authorities. With firm-level administrative data, this paper uses a spatial regression discontinuity design by comparing firm performance around the county border when one county is treated and the other is not. The results suggest significant improvements at the firm level after the hierarchy reform, including sales, employees, profits, and firm entry. Event studies on other county-level outcomes also suggest similar results in that GDP, fiscal income, household savings, and agriculture products all increased after the reform. Overall, this paper suggests that the hierarchy reform has improved overall social welfare, accelerated economic development, and expedited urbanization in China.

Keywords: hierarchy structure, centralization, urbanization, economic growth

1 Introduction

The design of the political hierarchy structure has long fascinated economists and policy-makers since it exerts a considerable influence on the development of urbanization and regional development (Henderson and Becker 2000). Extant research mainly focuses on the ramifications of democratization and favoritism. For instance, Burgess et al. (2015) demonstrate that democratization serves to decrease ethnic favoritism, thereby enhancing economic growth. Hodler and Raschky (2014) employ global data to illustrate that countries with weak political institutions and poorly educated citizens are more susceptible to regional favoritism. In addition to democratization and favoritism, centralization is a another pivotal feature of political institutions. The centralization or decentralization of the government's hierarchical structure would have a significant impact on the local economy's performance, and this paper wants to study this effects by exploiting a hierarchy reform policy in China between 2000 and 2013.

China's hierarchy structure has some specialties that makes it convenient to analyze the centralization and decentralization issues. In China, there is a multi-layer political hierarchies with a central government and various levels of local government. For the county level government, there are three distinct types of county-level administrative divisions: county, county-level city, and districts. Counties are typically the most rural and underdeveloped regions, as well as the most autonomous. When a county satisfies a set of conditions, such as a minimum GDP and population requirement, it may initiate a hierarchy reform (with the approval of the central government) and transform from county to district or from county to county-level city. In the first case, county level government will lose it's independence and become more centralized, and this paper choose this type of hierarchy reform to be the policy of interest. I analyzed the influence of political centralization on local economic performance by comparing its various manifestations.

First, I use an administrative firm survey data and find that after the political hierarchy change, firms become larger in terms of sales, employees, and profits. This means that the incumbent firms are better off after this policy. But to understand the whole picture of a firm's activities, it is also important to understand the firm entry. The full firm entry data is too large to geocode. Therefore, I draw a 5% random sample from

the original data and find that in every 5 km by 5 km block, the number of new firms increases by 3.5 after the hierarchy change. If the random sample is representative, then the number of new firms will increase by 70 in each block in total.

However, firm activities cannot fully characterize the economy, and the previous spatial regression continuity does not show the long-term trend after the policy. Therefore, I use the event study approach to analyze county-level data and find that county-level GDP, fiscal income, and household savings all increase after the hierarchy change. This effect is still persistent even after 18 years. One might argue that GDP may be manipulated and overestimated. Then I also use nighttime light as a proxy for the GDP, and find that the trend after the treatment is similar to the GDP. But the trend of nighttime light before the hierarchy change is smoother than the GDP, which indicates that government may overheat the economy or over-report the GDP prior to the treatment in order to be eligible for the hierarchy change.

This paper contributes to a significant range of literature, particularly the centralization and decentralization debate. Numerous studies of organization hierarchies theoretically analyzes the trade-off between vertical and horizontal controls (Calvo and Wellisz 1978; Tirole 1986; Qian 1994; Chen and Suen 2019), and there are also a number of studies modeling the tradeoffs between centralization and decentralization (Oates et al. 1972; Seabright 1996; Besley and Coate 2003; Cai and Treisman 2004). However, empirical evidence analyzing the political organization structure is limited and mixed. In the context of China, even though the government organization is thought to be an important factor of economic growth in theory (Maskin et al. 2000; Qian et al. 2006; Xu 2011), there is still no consensus on what the effect of political structure is. For example, Li et al. (2016) demonstrate a decline in GDP per capita as a result of government centralization and flattening. However, Young (2000) argues that decentralization will hurt the domestic market by increase local government's rent-seeking behaviors. Poncet (2003), Bai et al. (2004) and Cai and Treisman (2006) have probed the pitfalls of decentralization in China in greater detail. Tang and Hewings (2017) find that the incorporation of counties into prefectures increased the economic growth of these counties over time. This paper, however, differs from theirs, since I focus on the hierarchy change from county to district, and use rich administrative data with spatial regression discontinuity design and event study

approach.

The remainder of this paper is organized as follows. Section 2 discusses the institutional background. Section 3 discuss the data we use; section 4 presents the research design and the empirical findings; and section 5 concludes.

2 Institutional Background

There are four subnational administrative divisions in China: province, prefecture, county, and township. By the end of 2021, China has a total of 34 provinces, 333 prefectures, and 2843 county-level divisions. The average number of prefectures within a province is 12, while the average number of counties within a prefecture is 8. Township governments, which are at the lowest level, have limited economic decision-making autonomies and are therefore not encased in this study. Comparable to counties in the United States, the county-level division is the basic unit of China's local government hierarchy. As county-level governments play a vital function in China's political and economic processes, the distribution of governance responsibilities and powers is essential to comprehending Chinese local governments.

The focus of this paper is the relationship between prefecture and county governments. A province consists of multiple prefecture-level divisions, and each prefecture-level division consists of multiple county-level divisions. This structure is depicted in the Appendix Figure 3. Between the provincial and county governments, there is no direct relationship (Lou and Wang 2008). Prefecture-level governments play an intermediate role between a province and its component county level divisions.

There are three distinct divisions at the county level: county, county-level city, and districts. Figure 4 in the appendix characterizes an example of the hierarchy structure. All counties and county-level cities are nearly autonomous, and county governments have decision-making authority over the development of their own counties. The prefecture's government possesses no authority over the counties. In other words, it does not constitute a formal level of local administration (Chung and Lam 2004). Counties under any prefecture are able to make independent economic and political decisions regarding their own affairs. For instance, they can decide where public funds will be allocated and what public goods

will be provided, as well as local taxes and subsidies and the location of newly opened state-owned enterprises. Governments at the prefecture-level have limited authority to intervene with county-level governments because, by law, prefectures can only relay information to the counties and supervise them on behalf of provincial governments. Prefectures are not permitted to determine how counties spend their public funds, transfer revenues between counties, or approve the establishment of new state-owned enterprises(Bo and Cheng 2021).

In contrast, district governments do not have the authority to make decisions. In general, the fiscal, administrative, and personnel decision-making authorities are transferred to prefecture governments (Shi et al. 2009). First are the authorities over public finance revenue and expenditures. On the revenue side, prefecture level governments are now sharing tax revenue with districts, forcing districts to rely more on transfer payments from higher levels of government. On the expenditure side, prefecture-level governments assume increased responsibility for public expenditure. Second, administrative authority over social and economic undertakings is taken over by the prefecture government. For instance, the establishment of a state-owned enterprise can no longer be determined by the districts in which it is located. Third, counties also lose the powers to appoint, reward and punish local officers if they becomes districts. District officers must therefore follow orders from cities at the prefecture-level.

In the process of economic development, however, not all counties grow at the same rate, so the status of a county is not perpetual. Some counties have a progressively higher GDP and population than others and become so urbanized that it is no longer reasonable to be considered as counties. Therefore, beginning in 1976, the central government decides to establish new cities and districts at the county level. The defining characteristic of this reform is that counties do not grow in size and do not take possession of additional landmass. The only difference is the structure of the political hierarchy. Figure 5 in the appendix depicts the number of new county-level cities and districts that are created each year. Before 1998, there was a local government "upgrade" craze. On average, more than 20 county-level cities were created each year, whereas only a handful of districts were created. However, this policy was halted by the central government in the 1990s due to the fact that too many counties sought urban designations and "upgrades" despite the reality

that some counties did not satisfy the necessary criteria. The period of stagnation lasted approximately 10 years. In principle, local counties are not permitted to apply for city improvements during this period (but we can still see a few exceptions from the figure). After 2010, the central government decides to restart the political hierarchy reform, but in the early years, counties are hesitant to implement this policy. The number of new districts and cities at the county level did not begin to hike until 2014.

This policy shock provides us an excellent opportunity to understand the political hierarchy reform because, in the previous and subsequent periods when counties could choose to be county-level cities or districts, this would necessitate extensive bargaining and lobbying, and create an endogeneity problem. During this period of stagnation, however, there is no need to ruminate about the complex endogenous decision of whether to become a county-level city or district, as all counties must be districts in order to be urbanized. Therefore, this paper will mainly focus on this stagnation period and also shortly after the stagnation.

3 Data

3.1 Political hierarchy reform data

All the policy hierarchy reform data comes from the Ministry of Civil Affairs which has a website to publish all the county-level hierarchy changes every year. This information gets published between 2000 and 2021. However, considering that one of our firms level data is only available between 1998 and 2013, and also the endogeneity reason aforementioned, I choose to focus on the period between 2000 and 2013, and analyze the hierarchy change during this period.

In total, there are 61 counties level divisions that have experienced the political hierarchy reform during this period, and I take these counties as the treated counties. I then selected the contiguous counties that are directly connected with these treated counties as our control group. In total, 290 counties have experienced this political hierarchy reform or are contiguous to reformed counties, accounting for 10% of Chinese county-level divisions. Figure 6 in the appendix depicts the geographic distribution of the counties that become districts during this period. The cyan areas represent the treated counties, which become

districts, whereas the orange areas represent the contiguous county-level division.

3.2 Firm level data

3.2.1 Annual Survey of Industrial Enterprises

In this paper, I have used two firm-level administrative data. The first one is the Annual Survey of Industrial Enterprises (ASIE), which is collected by the national bureau of statistics. This is an administrative data set containing all large industrial firms in China between 1998 and 2013. Industrial firms include manufacturing firms, mining firms, and electricity, gas and water firms. This paper will only use the manufacturing firms for two reasons. First, more than 90% of the industrial firms are manufacturing firms, and this is good coverage of the whole data set. Second, the mining, electricity, gas and water firms are usually state-owned companies and have very different characteristics compared to the average manufacturing firms.

In general, this ASIE data set is a large unbalanced administrative data set. There are more than 4 million firm-year observations, from more than 1 million unique firms. This paper geocodes the firms' address and get the latitude and longitude for the further spatial regression discontinuity analysis. The variables of interest include sales, employees, production, sales cost, administrative expenses, profits, total assets, and the distance between the firm and the border. The descriptive statistics can be seen in the appendix table 3. Here I drop all the data that the minimum sales, employees, production, assets, or costs are negative. I also drop the data that the distance from the border to the firm is larger than 30 km. The final number of observations that are available is approximately 1.2 million.

3.2.2 Firm registration data

The second firm-level data set is the firm registration data, which is another administrative data collected by the State Administration for Market Regulation. It includes all the firms that are legally registered in China starting from 1949 when the people's republic of China was established. This data set provides three unique advantages compared to the previous ASIE data set. First, it's a large data set starting from 1949 and keeps updating every

day. Second, This firm registration data contains both large and small firms, and both industrial firms and service firms. Therefore, it could help us to understand the whole picture of China's firm activities after the political hierarchy reform. Third, this firm registration data also allows us to understand the firm entry behaviors, while the ASIE data only contains the incumbent.

The firm registration data is a huge data set containing approximately 260 million firms between 1949 and 2021, and to apply the spatial regression discontinuity method, it's essential to first geocode these firms, which turns out to be impossible for this paper since the map API firms ask for several thousand pounds to geocode 260 million addresses. Therefore, two strategies are used here to circumvent this problem. First, I sum up the total number of new firms in every county by year and use the aggregate level data as the outcome variable. Second, I limit the time span of these firms and only focus on the firms that are established between 2000 and 2013, which is the same as the period of political hierarchy reform data. This process could drop 184 million firms, but the remaining 74 million is still too large (and expensive) to handle. Then I select the firms that are either located in the treated counties or the contiguous counties, and this would give me 35,595,654 firms. Finally, I draw a 5% random sample from the remaining firms and get 1,779,783 firms.

3.3 County level data

This paper also collected the county-level economic data from the statistics yearbook published by the national bureau of statistics between 2000 and 2020. Unfortunately, there are many missing values and some variables are not collected in some years. Among these 2843 county-level divisions, only 2545 of them have a significant amount of non-missing data and can be transformed into panel data. The variables of interest include the county level GDP, GDP by sector, fiscal income, household savings, grain output, oil crops, and hospital beds. The descriptive statistics for these county-level variables can be seen in table 4 in the Appendix.

The previous statistics yearbook data covers almost 90% of the county-level regions. However, there are still 10% missing regions, and even for these 2545 counties, the GDP data is only available for 90% of the time. To overcome this problem, I also use the

following satellite data to proxy the county-level GDP. The original raster data is provided by Chen et al. (2020) and it is a global NPP-VIIRS-like nighttime light data that has been cross-sensor calibrated. I then obtain the average, minimum, maximum, aggregate and variance of the night light intensity in each county-level region.

4 Research Design and Empirical Finding

4.1 Spatial regression discontinuity

The main identification strategy is to compare the firms' activities in the treated regions with the firms in the contiguous counties. This paper follows Michaels et al. (2021) and Gelman and Imbens (2019) by applying the following Spatial Regression Discontinuity Design.

$$y_{i,t} = \beta_1 Reform_{j,t} + \beta_2 Dist_{i,t} + \beta_3 Dist_{i,t} \times Reform_{j,t} + \beta_4 \textbf{Controls} + \textbf{Fixed effects} + \epsilon_{i,t}$$

Here, $y_{i,t}$ measures various firm-level outcomes as described in the data section and appendix. $Reform_{i,t}$ is the main variable of interest and is a dummy variable indicating whether this county has experienced the political hierarchy reform. $Dist_{i,t}$ measures the distance in kilometers between the firm and the county border. **Controls** is a vector of control variables, including the age of the firm and also polynomial terms of the distance. **Fixed effects** is a vector of fixed effect, including city×year fixed effect and firm fixed effect.

The baseline analysis uses firms within 30 km of the border in the treated and contiguous counties. I use this cutoff because almost all firms in the treated counties are within 30 km of the border. In the robustness check section, I also present alternative cutoffs using 10 km and 20 km, and the results are similar to the baseline analysis.

I also cluster the standard errors following the ideas of Michaels et al. (2021) and Bleakley and Lin (2012), to account for the spatial correlation. The standard errors are clustered on 0.1° latitude × 0.1° longitude block, which is roughly a 10 km × 10 km block. The 10 km distance reflects the average distance between the firms and the border conditioning on the 30 km cutoff. Also, the size of the block is approximately 5% of a

median sized county level region. In the robustness check section, alternative size of the clusters is used and the results are basically unchanged.

4.1.1 Firm economic activities

Table 1 shows the results of the regression discontinuity. Here the variable of interest is the firm's sales and the number of employees. I take logs of both variables to make the coefficient semi-elasticity. Column (1) and column (4) is the result without any controls and fixed effects. When a county becomes a district, firms' sales increase by 26.9% and this term is economically and statistically significant. However, for the employees, it seems that when a county experienced a political hierarchy reform, the number of employees is not much affected. Then for column (2) and column (5), I add more control variables, and the result is quite stable. The sign and the magnitude of the coefficients do not change at all. Finally, for the columns (3) and (6), I add the prefecture $\times$ year fixed effect. For the sales, the sign is still positive, but the magnitude drops to 0.06, which means firms' sales will increase by 6% when the county becomes a district. For the employees, the coefficient becomes significant, and the number of employees will increase by 4.5% after the political hierarchy reform.

From the table 3 in the appendix, I also presented the results using different outcome variables. In column (1), instead of using sales, I directly use the production data. Compare to the effects on the sales, the effects on the production have the same sign and very similar magnitude. In column (2), the total assets of the firms in the treated counties also increase by 4% after the political hierarchy change. From column (3) and column (4), the sales cost and administrative expenses increased by 5.6% and 4.3% respectively. And from column (5) the firm's profits increase by 5.5% after the hierarchy change. All this evidence indicates that the political hierarchy reform makes firms to grows bigger. However, we still don't know whether this growth of firms truly reflects the prosperity of the local economy. One potential argument is that even though the existing firms grow bigger, it is possible that the number of new firms decreases, which makes the economy worse off. This argument will be tested later using the firm entry data.

Table 1: the effect of hierarchy change on firm's economic performance

	(1) sales	(2) sales	(3) sales	(4) employees	(5) employees	(6) employees
treat	0.264***	0.274***	0.063***	-0.117	-0.114	0.045***
	(4.35)	(5.70)	(4.11)	(-1.13)	(-1.20)	(3.81)
dist	-0.006	-0.010	0.013**	0.002	0.019	0.009**
	(-1.10)	(-0.36)	(2.29)	(1.07)	(0.71)	(2.29)
treat × dist	0.001	-0.001	-0.000	0.017**	0.017**	0.001
	(0.16)	(-0.20)	(-0.03)	(2.16)	(2.40)	(1.30)
Observations	1,242,415	1,241,625	1,161,359	1,439,334	1,438,300	1,313,125
Controls	No	Yes	Yes	No	Yes	Yes
prefecture × year FE	No	No	Yes	No	No	Yes
Firm FE	No	No	Yes	No	No	Yes
Distance	< 30km	< 30km	< 30km	< 30km	< 30km	< 30km

Note: I take log of all the outcome variables. Control variables include the age of the firm and also polynomial terms of the distance. The standard errors are clustered on 0.1° latitude × 0.1° longitude block, *** p<0.01, ** p<0.05, * p<0.1

4.1.2 Firms entry

Firm sales and employees only capture the intensive margin of the firm's activities. Another important part of a firm's activity is the firm entry, which captures the extensive margin. Table 2 shows the result of the effect of political hierarchy reform on firm entry. Here to measure the firm entry, I use an arbitrary 0.05° longitude × 0.05° latitude block, which is approximately 5 km by 5 km. Then I count the number of new firms in this block this year and the average distance between these firms in this block to the nearest border. Columns (1) to (3) show the effects of the hierarchy change on the number of new firms in the 0.05° longitude × 0.05° latitude block. In columns (4) to (6), I change the size of the block to 0.1° longitude and 0.1° latitude. Columns (1) and (4) do not add any control variables and fixed effects. Then columns (2) and (5) add control variables, and the control variable is the polynomial terms of the distances. Columns (3) and (6) further add the time fixed effect. I also add the condition that the average distance of these blocks is smaller than 30 km to make it comparable to the previous results.

Column (1) is the simplest OLS result without any controls, and it seems that political hierarchy reform does not change the firm entry behavior. This non-significant result is still positive for the policy because we know from the previous results that political hierarchy reform can increase a firm's sales and employees. As long as the firm entry

effect is non-negative, then the total effect would be positive. In columns (2) and (3), when I add controls (and time fixed effect), the firm entry effect becomes positive. In columns (4) to (6), I use a larger cell to measure the firm entry, and the result is again highly significant, which shows that more firms are entering the market after the hierarchy change. To interpret the coefficient, we need to bear in mind that this is a 5% random sample from the total firms, and the outcome variable is the number of new firms in a block. If this random sample is representative, then we should multiply the coefficient by 20 to get the total effect. If we look at column (3), then the hierarchy reform will bring approximately 70 new firms per block.

Table 2: the effect of hierarchy change on the firm entry

	(1)	(2)	(3)	(4)	(5)	(6)
	$0.05° \times 0.05°$	$0.05° \times 0.05°$	$0.05° \times 0.05°$	$0.1° \times 0.1°$	$0.1° \times 0.1°$	$0.1° \times 0.1°$
treat	1.262	3.838**	3.533*	12.778***	17.211***	15.794***
	(0.63)	(2.11)	(1.93)	(3.58)	(4.70)	(4.31)
dist	-0.172**	0.052	0.047	-0.640***	-0.430	-0.445
	(-2.33)	(0.13)	(0.11)	(-9.47)	(-1.34)	(-1.39)
treat × dist	-0.441**	-0.819***	-0.818***	-1.892***	-2.466***	-2.487***
	(-2.00)	(-3.64)	(-3.65)	(-3.47)	(-4.44)	(-4.48)
Observations	89,533	89,533	89,533	35,890	35,890	35,890
Controls	No	Yes	Yes	No	Yes	Yes
year FE	No	No	Yes	No	No	Yes
Distance	< 30km	< 30km	< 30km	< 30km	< 30km	< 30km

Note: The outcome variable is the number of new firms in each block. Control variables include polynomial terms of the distance. *** p<0.01, ** p<0.05, * p<0.1

4.1.3 Further discussion

In the appendix, I carry a list of robustness checks and the results are all similar to the baseline analysis. First, instead of using 30 km as the cutoff distance, I also tried 10 km and 20 km, and the results are presented in Table 4. When we choose a smaller distance bandwidth, we are in essence choosing a smaller sample and the firms in this sample are geographically more close to each other. Column (1) and column (2) is the sample within 10 km from the border, and column (3) and column (4) is the sample within 20 km from the border. From this table, we can see that the coefficient is still positive and significant. However, the magnitude is smaller in this case. Previously, the hierarchy change will lead to a 6.3% increase in sales and a 4.5% increase in the employees.

However, within 10 km distance, the effects become 4.9% and 4% respectively, and within 20 km distance, the effects are 4.9% and 3.6% respectively. Second, instead of using 0.1° longitude × 0.1° latitude block, I also tried the 0.05° longitude × 0.05° latitude block, which is approximately 5 km by 5 km. From table 5 in the appendix, it is shown that the magnitude of the coefficient is not changed, but the standard error becomes smaller. In table 7, I also try to add other county-level control variables, such as the county level GDP and population. I didn't add them in the baseline regression because county-level data have missing values. If we add both county-level GDP and population, the number of observations drops from 1.16 million to 0.6 million. Actually, these county-level variables can be partly absorbed by the prefecture × year fixed effect in the baseline regression. Therefore, we should not expect a huge difference if we add these county-level control variables. This is indeed the case in Table 7, and most of the coefficients are not changed in terms of magnitude and significance. The only difference is that profits are no longer significant.

Another interesting question to ask is whether firms become more productive after the hierarchy change. Here I use the same set of outcome variables and divide them by the number of employees to get a rough estimate of the labor productivity. In table 6, the coefficient for all different variables is not significantly different from zero. In other words, labor productivity is not likely to be affected by this hierarchy reform.

4.2 Event study

There are still some other problems with the previous analysis. The firm sales and employees data only cover large industrial firms, and small firms or non-industrial firms are not included in the ASIE data set. That's why I also use the firm entry data because this administrative data set contains all firms that are legally registered in China no matter their type and size. However, the second data set introduces a new problem. Since the administrative firm entry data set contains more than 35 million firms between 2000 and 2013 in the treated and contiguous counties, which makes the geocoding very expensive. Therefore, I have to draw a 5% random sample from this data set and analyze these randomly selected firms. Whether the result from this 5% random sample can be generalized to the whole sample is still questionable. A third problem is that there are

many other dimensions to measure the local economic performance other than the firms' activities. Firm sales and firm entry are important parts of the economy, but the GDP, household wealth, and public services are also very important.

To solve the previous problem, and also to get a more complete picture of the effects of the reform, I use the aggregate county level data to implement the event study design. The model specification follows Cesarini et al. (2017), and I normalize the time of the hierarchy reform to t = 0

$$y_{i,t} = \beta_t Reform_{i,0} + \beta_1 \textbf{Controls} + \textbf{Fixed effects} + \epsilon_{i,t}$$

The term $Reform_{i,0}$ is a dummy variable indicating whether this county has political hierarchy change. **Controls** include a list of control variables, including the population and the the area of this county, the distance between this county and the capital of the prefecture, and the distance between this county and the capital of the province. **Fixed effect** is a prefecture × year fixed effect. The standard errors are clustered at the county level.

4.2.1 GDP and nighttime light

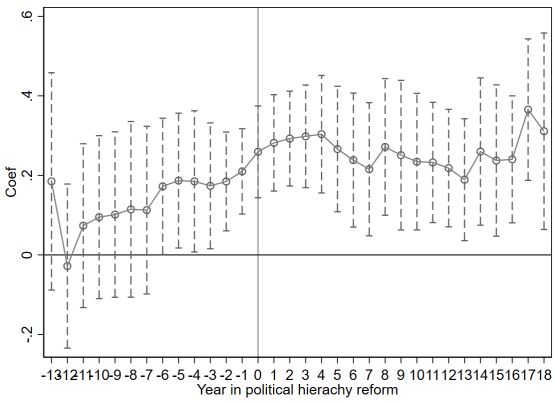

Figure 1: GDP

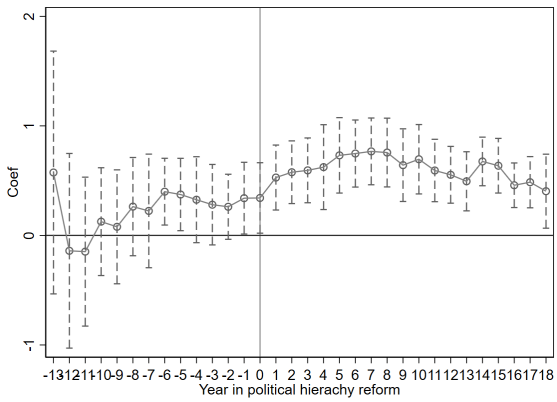

Figure 2: Night time light

Figure 1 shows the county-level GDP trend after the political hierarchy reform. After the hierarchy change, the county's GDP is consistently higher than the control group that has not experienced any hierarchy change. And it is also consistently higher than the previous years. However, as we can see from this figure, according to the GDP measure,

the county's GDP is already high even before the hierarchy change. The coefficient is significantly different from zero two years before the hierarchy change. There are several reasons to explain this difference. First, to be eligible for the hierarchy change, this county needs to satisfy a set of conditions, such as a high GDP and a high population requirement. Therefore, It is not surprising that the GDP is higher than the control group before the hierarchy reform. This GDP growth may be simply driven by the one time increase of government purchase or new construction projects that cannot last long. It's more important to see whether this county can sustain this high GDP after this hierarchy reform. According to this figure, this answer is yes, and counties can keep a high GDP in the following 18 years, with no evidence of any decreasing trend. In this sense, this political hierarchy reform indeed makes the county better off, or at least successfully finds these good counties. Second, to understand the GDP jump before the hierarchy reform, we could also consider government incentives. If the county government wants to upgrade, they have the incentive to manipulate the GDP data to persuade the central government to approve this upgrade. Therefore, the jump before the reform may be fake and overestimated, and the GDP after the hierarchy reform would be more reliable. If we believe in this point, then figure 1 would be even more positive, because it says that even if a county manipulates the GDP data to get the county upgrade, this county can still sustain a long time high GDP.

Another way to see the GDP or the overall economic performance is through the local government's fiscal income. In general, the higher the GDP, the more government can tax and the higher the fiscal income. However, the government has less incentive to over-report the fiscal income because a proportion of this income must be delivered to the central government. As we can see in Figure 7 in the appendix, the fiscal income increases just one year before the hierarchy reform, compared to the GDP increases five years before the hierarchy reform. The post-treatment trend for the fiscal income is similar to the GDP and the coefficient is significant for the next 17 years.

Nighttime light is another common way to measure economic performance, especially in developing countries where the data is less available and reliable. Interestingly, if we look at the nighttime light in figure 2. It seems that the nighttime light does not increase at all compared to the control group before the hierarchy reform. However, immediately

after the reform, the nighttime light becomes significantly higher, and this sustains for 18 years. Therefore, combining the GDP result with the fiscal income result and nighttime night result, it is possible that the GDP before the hierarchy reform is overestimated. A more detailed discussion of the fake GDP issue is beyond the scope of this paper. However, even though the pre-treatment GDP may not be that reliable, the post-treatment trends of the GDP, fiscal income, and nighttime light are consistent, and it allows us to argue that the political hierarchy reform indeed increases the local economic performance in the long run.

4.2.2 GDP by sector

Figures 8, 9, and 10 in the appendix show the effects of political hierarchy change on the primary, secondary, and territory sectors. For the primary sector, the GDP is higher than the control group before the political hierarchy reform is implemented. Immediately after the political hierarchy change, the primary sector drops and it drops even further after 14 years. In general, we can see a very clear decreasing trend in the primary sector. For the secondary and territory sector, there is a jump 2 years before the hierarchy change, which is not surprising because the county's GDP from the secondary and tertiary sector must be high in order to be eligible for a hierarchy change. After the hierarchy change, secondary and tertiary sectors are persistently high and the trend is quite stable. From these three figures, we can see that this policy is a successful way to accelerate urbanization, leading to a shrinking primary sector and expanding secondary and tertiary sectors.

Figures 11 and 12 in the appendix further elaborate on the pattern of the agriculture sector through the change in grain output and oil crops. There is a decreasing trend for the grain output after the hierarchy reform, even though this decrease is not significant until ten years later. The trend for the oil crops is also decreasing, and the coefficient is significantly smaller than 0 immediately after the hierarchy reform. The coefficient for the next 18 years is either significantly smaller than 0 or non-positive. In general, these two figures convey the information that the agricultural sector shrinks after the hierarchy reform, which is consistent with the previous graph of GDP in the primary sector.

Figures 13 and 14 in the appendix help us understand the secondary and territory sector, through the number of new enterprises and sole proprietorships in this county.

This firm entry data is directly from the State Administration for Market Regulation and does not subject to the local government's manipulation. The coefficient for the number of new enterprises is not significantly different from 0 before the hierarchy change, becomes positive immediately after the hierarchy change, and remains positive in the next 15 years. The coefficient for the number of new sole proprietorships does not respond to the hierarchy reform in the first few years. However, 4 years later, this coefficient becomes positive and remains significant for approximately 10 years. According to these two pieces of evidence, more firms are entering the market after the hierarchy reform, which is a strong indication the prosperity of an economy, especially in the secondary and territory sector.

4.2.3 Do people benefit from the hierarchy change

From the previous results, we observe a better economic performance after the political hierarchy change. However, it is not sure whether this economic growth leads to better household well-being. Here we directly show the household savings before and after the hierarchy change, to elaborate on whether they really become richer after this change. Figure 15 shows the household savings, and the coefficient is significantly higher than zero in the next 17 years after the hierarchy change.

We can also look at the public service provision to get a better sense of the local resident's well-being. In China, nearly all hospitals are public hospitals, and the number of hospital beds is a good indicator of the health service provision in this area. In figure 16 in the appendix, the number of hospital beds is nearly unchanged before and after the hierarchy change. We can only see two significant positive coefficients in the 14th and 15th years, which may be simply driven by some outliers. Therefore, in terms of the health sector, the hierarchy reform has almost no effect, which indicates that the public service provision of these counties is still lagging behind others.

5 Concluding Remarks

This paper investigates the effects of political hierarchy change on local economic performance at the micro-firm and county levels. Economists have long been fascinated

by political hierarchy structure. Numerous theories have been proposed to explain the political hierarchy, but empirical evidence is inconsistent. I primarily focus on the shift from county to the district between 2000 and 2013 in China. Local governments become more urbanized as a result of this change in hierarchy, but they lose autonomy and become more centralized. I use spatial regression discontinuity to examine how a firm's activities are affected by a change in the hierarchy and discover that, in general, the performance of firms improves after a hierarchy change. Then, utilizing an event study to analyze county-level data, it is concluded that the county as a whole is also better off after the hierarchy change.

For the firm-level analysis, high-quality administrative data have been applied to understand the behavior of firms as measured by sales, employees, production, cost, and profit. After the restructuring of the hierarchy, all outcomes increase simultaneously, indicating the expansion of the firms. The growth of firms is accompanied by an increase in the number of firms entering the market, indicating that the market has become more competitive as a result of the change in the hierarchy and that the growth of firms is not solely driven by the redistribution of revenue.

For country-level analysis, the data from statistical yearbooks and satellites have been adopted. The event study reveals an increase in county-level GDP and nighttime light following the hierarchy change. Additionally, household savings and government fiscal income start rising, indicating that local residents and government are indeed becoming wealthier. The contraction of the primary sector and expansion of the secondary and tertiary sectors indicate that the county is indeed becoming more urbanized. However, there is no evidence that the health service has changed. Even though the county is more urbanized than others, the public service is still lagging.

Our findings suggest that a change in the political hierarchy can improve the local economy. Businesses, households, and local governments can all benefit from this change in hierarchy. An intriguing question to ask is what mechanism is responsible for this difference. A comprehensive analysis of the mechanism is beyond the scope of this paper, but I find that the policy does not affect labor productivity. Additionally, we must interpret the result with caution, particularly in terms of its external validity. If all counties become districts, for instance, it is possible that we will not be able to observe such a large and

significant improvement in the local economy. Even with the aforementioned caveats, this policy demonstrates that the central government can play a role in urbanization planning, and this lesson may be beneficial for other developing countries.

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7 Appendix

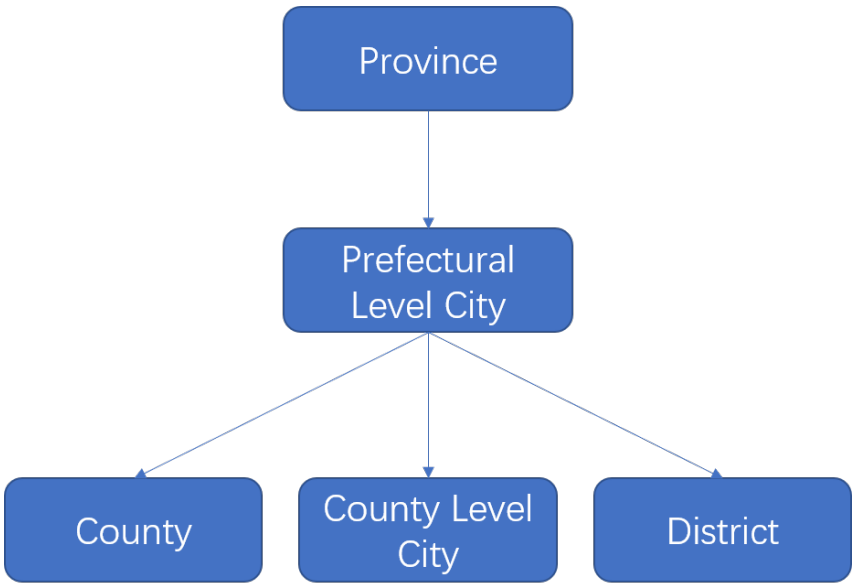

Figure 3: simplified administrative division stricture in China

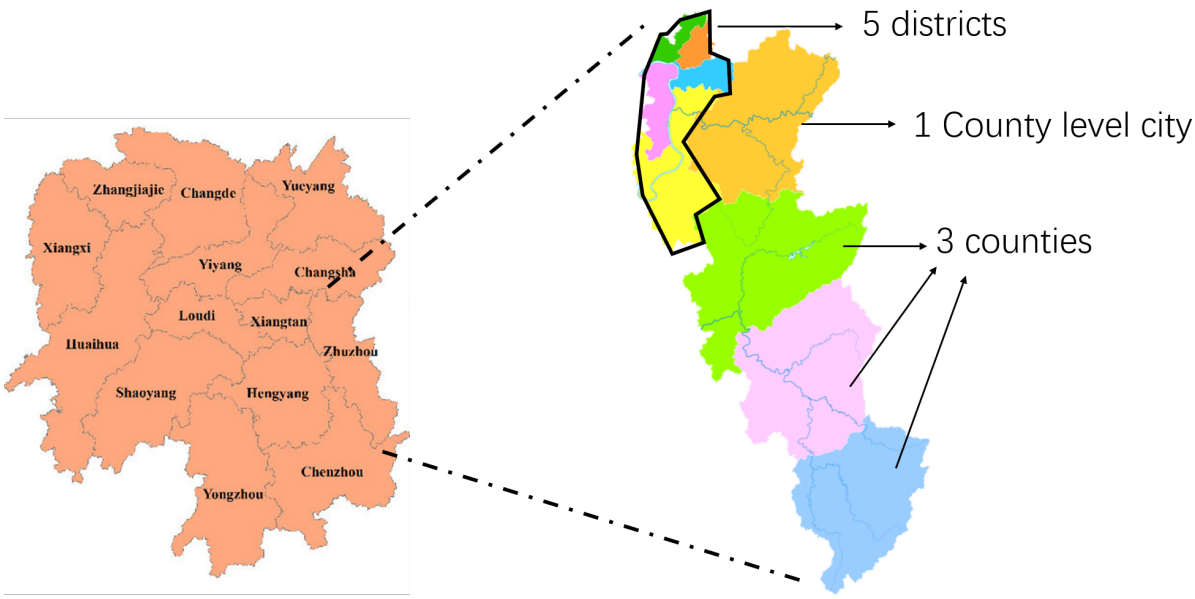

Figure 4: an example of province-prefecture-county structure

Note: The province of Hunan consists of 13 prefecture-level divisions. Zhuzhou is one of these prefecture divisions; it is comprised of 3 counties, one county-level city, and 5 districts.

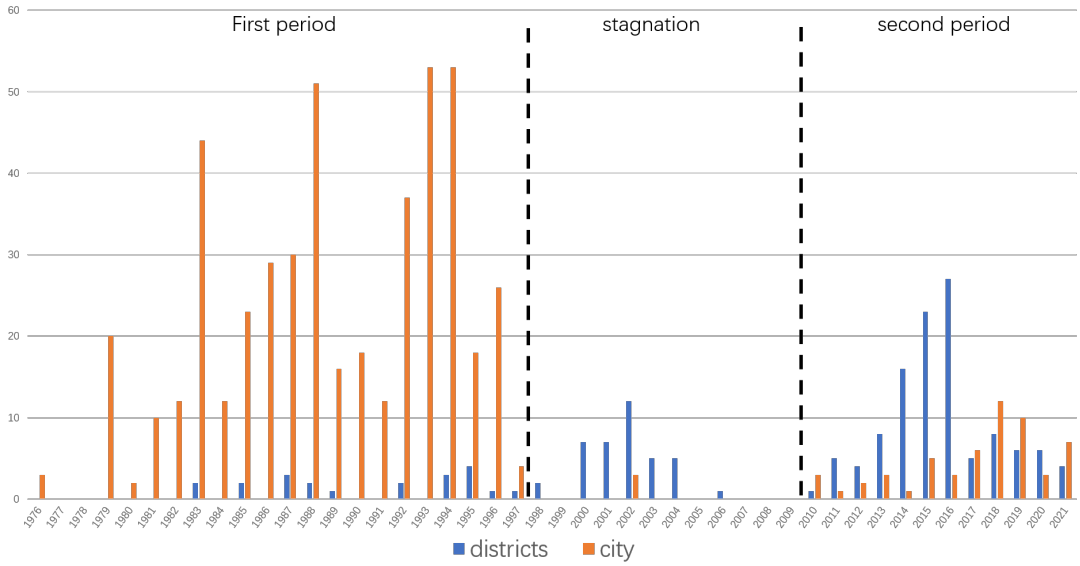

Figure 5: the number of new districts and new county level cities

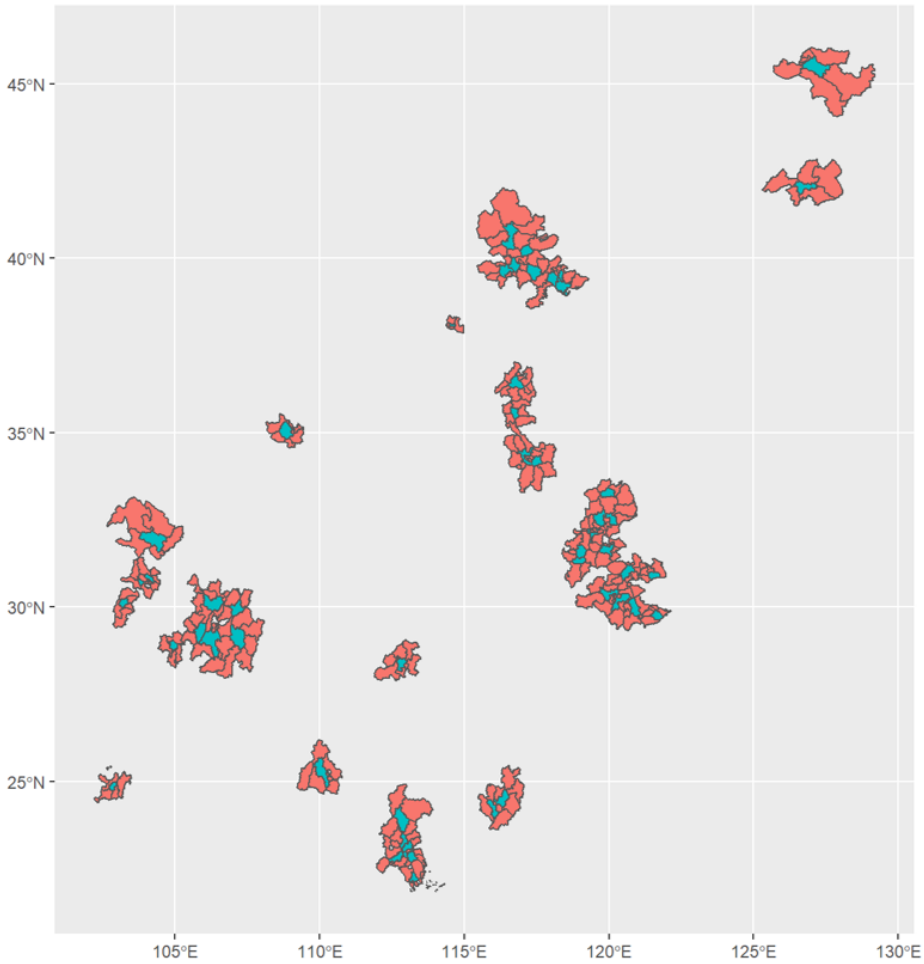

Figure 6: treated counties and it's contiguous county level divisions

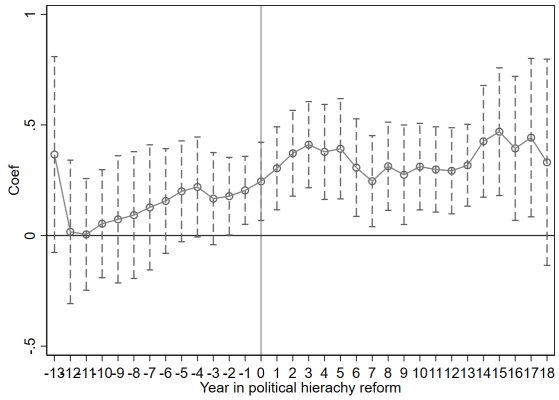

Figure 7: fiscal income

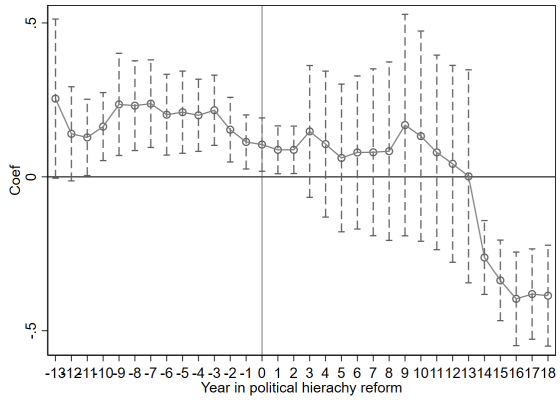

Figure 8: primary sector

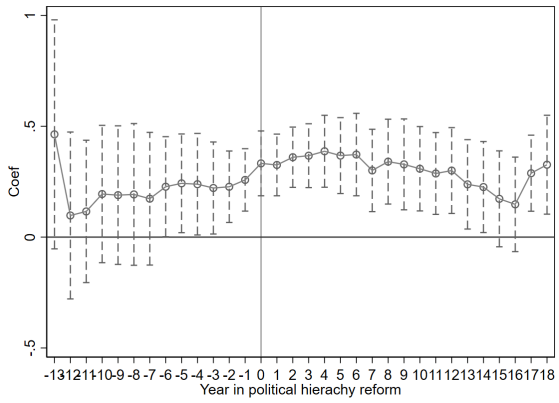

Figure 9: secondary sector

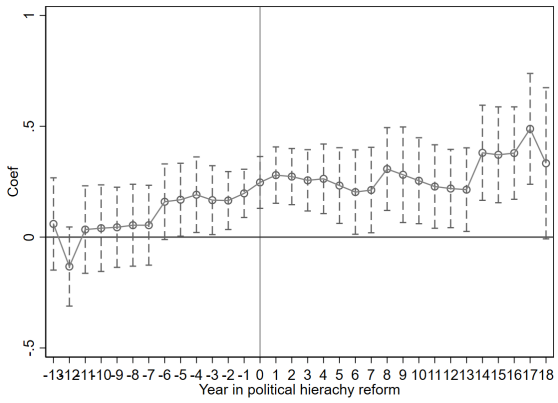

Figure 10: tertiary sector

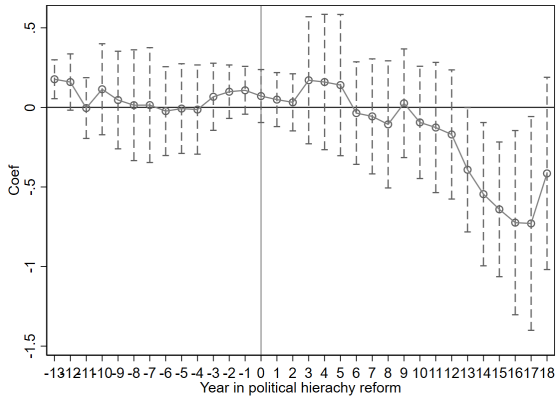

Figure 11: grain output

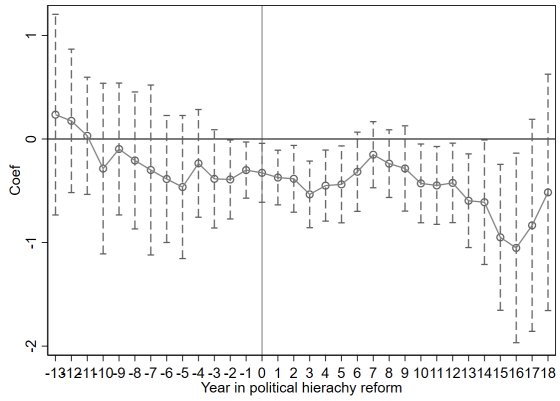

Figure 12: oil crops

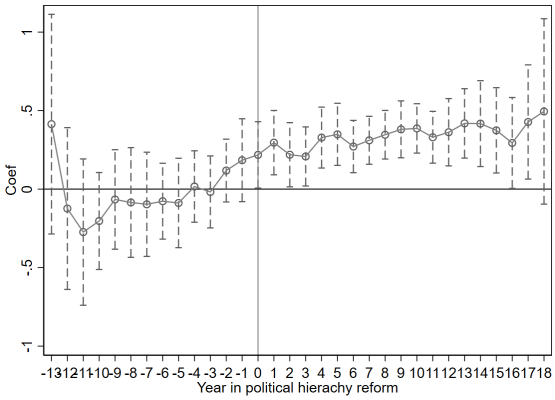

Figure 13: new enterprises

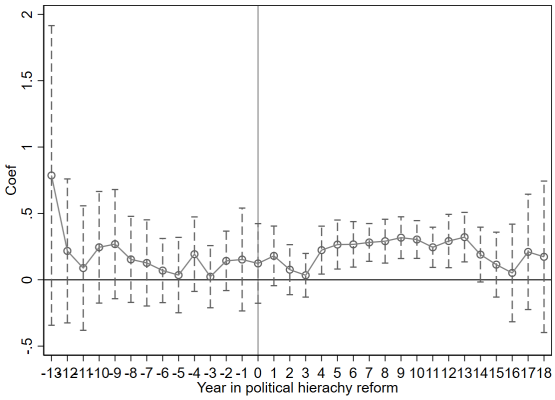

Figure 14: new sole proprietorship

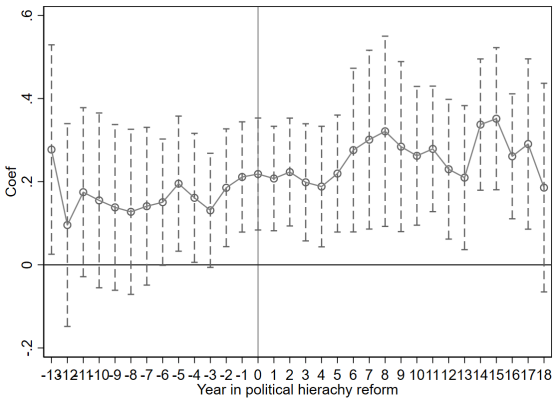

Figure 15: household savings

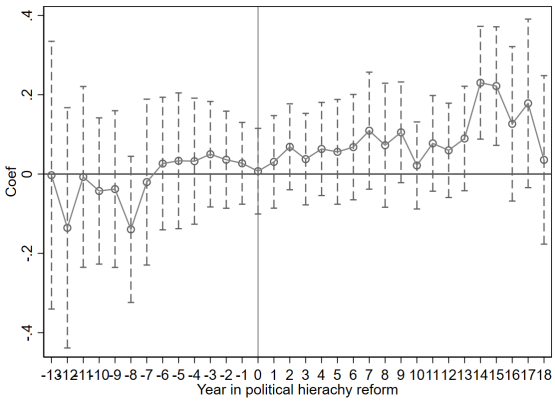

Figure 16: hospital beds

Table 3: descriptive statistics for the ASIE data

	Mean	SD	Min	Max	N
sales	135,688.97	1,286,668.51	0	259502670	1,242,415
employees	257	1,001.94	0	147,182	1,439,334
production	138,398.64	1,295,499	0	259502670	1,242,489
asset	125,892.38	2,185,590.54	0	949758335	1,521,045
sales cost	115,476.90	2,379,845.05	0	2400618552	1,351,095
administrative expenses	5,753	44,580.92	0	7,200,000	1,350,709
profit	7,596.18	115,310.20	-7,625,896	31,286,492	1,350,885
distance	11.81	8.51	0	30	1,521,133

Table 4: descriptive statistics for the county level data

	Mean	SD	Min	Max	N
GDP	1,360,650.66	2,457,439.13	3,102	92,703,100	47,944
primary sector GDP	167,024.11	174,586.71	1	2,096,700	47,801
secondary sector GDP	630,900.91	1,341,528.04	1	58,284,300	48,014
tertiary sector GDP	539,048.40	1,151,359.84	1,212	34,635,800	47,475
fiscal income	83,743.83	197,797.45	6	6,729,838	49,830
household saving	829,578.62	1,444,931.24	0	32,551,489	45,233
grain output	237,483.23	277,396.38	0	3,640,712	46,325
oil crops	13,756.27	25,322.67	0	896,301	47,033
hospital beds	1,417.09	1,433.25	8	16,669	45,940
nighttime light density	3,115.50	6,913.43	0	212,112.06	60,879
new enterprises	2,975.40	5,561.56	0	217,242	58,120
new sole proprietorship	1,857.94	3,516.84	0	198,127	58,120

Table 5: with different outcome variables

	(1) production	(2) asset	(3) sales cost	(4) admin expenses	(5) profit
treat	0.066***	0.044***	0.056***	0.043**	0.055**
	(4.25)	(3.46)	(3.76)	(2.34)	(2.14)
dist	0.014**	0.009**	0.010*	-0.002	0.009
	(2.56)	(2.05)	(1.96)	(-0.30)	(0.95)
treat×dist	-0.000	0.000	0.000	0.001	0.001
	(-0.02)	(0.28)	(0.12)	(0.60)	(0.52)
Constant	10.233***	9.976***	9.922***	7.229***	6.274***
	(81.44)	(77.36)	(73.75)	(71.80)	(36.07)
Observations	1,161,439	1,394,268	1,262,203	1,261,682	1,039,339
R-squared	0.825	0.870	0.824	0.812	0.753
Controls	Yes	Yes	Yes	Yes	Yes
prefecture × year FE	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes
Distance	< 30km	< 30km	< 30km	< 30km	< 30km

Note: I take log of all the outcome variables. Control variables include the age of the firm and also polynomial terms of the distance. The standard errors are clustered on 0.1° latitude × 0.1° longitude block, *** p<0.01, ** p<0.05, * p<0.1

Table 6: with different cutoff distances

	(1) sales	(2) employees	(3) sales	(4) employees
treat	0.049***	0.040***	0.049***	0.036***
	(2.76)	(2.86)	(3.18)	(2.92)
dist	0.040	0.019	0.022**	0.010
	(1.62)	(1.01)	(2.48)	(1.38)
treat×dist	0.002	0.003	0.000	0.003*
	(0.61)	(1.16)	(0.09)	(1.73)
Constant	10.291***	4.902***	10.243***	4.874***
	(91.13)	(65.07)	(68.11)	(49.30)
Observations	575,151	656,643	909,473	1,032,307
R-squared	0.839	0.832	0.828	0.827
Controls	Yes	Yes	Yes	Yes
prefecture × year FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Distance	< 10km	< 10km	< 20km	< 20km

Note: I take log of all the outcome variables. Control variables include the age of the firm and also polynomial terms of the distance. The standard errors are clustered on 0.1° latitude × 0.1° longitude block, *** p<0.01, ** p<0.05, * p<0.1

Table 7: with different clusters

	(1) production	(2) asset	(3) sales cost	(4) admin expenses	(5) profit
treat	0.066***	0.044***	0.056***	0.043***	0.055**
	(5.09)	(4.01)	(4.49)	(2.79)	(2.44)
dist	0.014***	0.009**	-0.002	0.010	0.009
	(2.73)	(2.25)	(2.04)	(-0.32)	(1.02)
treat×dist	-0.000	0.000	0.000	0.001	0.001
	(-0.03)	(0.32)	(0.16)	(0.64)	(0.56)
Constant	10.233***	9.976***	9.922***	7.229***	6.274***
	(85.71)	(78.15)	(76.97)	(73.62)	(37.67)
Observations	1,161,439	1,394,268	1,262,203	1,261,682	1,039,339
R-squared	0.825	0.870	0.824	0.812	0.753
Controls	Yes	Yes	Yes	Yes	Yes
prefecture × year FE	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes
Distance	< 30km	< 30km	< 30km	< 30km	< 30km

Note: I take log of all the outcome variables. Control variables include the age of the firm and also polynomial terms of the distance. The standard errors are clustered on 0.1° latitude × 0.1° longitude block, *** p<0.01, ** p<0.05, * p<0.1

Table 8: outcome variable per worker

	(1) sales	(2) production	(3) assets	(4) sales cost	(5) admin cost	(6) profits
treat	0.011	0.015	0.006	0.008	-0.009	-0.005
	(0.68)	(0.92)	(0.45)	(0.47)	(-0.52)	(-0.16)
dist	0.005	0.007	0.001	0.001	-0.007	0.005
	(0.93)	(1.19)	(0.24)	(0.16)	(-1.23)	(0.42)
treat_dist	-0.001	-0.001	-0.001	-0.001	0.001	0.002
	(-0.37)	(-0.56)	(-1.04)	(-0.38)	(0.43)	(0.78)
Constant	5.312***	5.317***	5.094***	5.029***	2.316***	1.367***
	(56.63)	(56.92)	(48.52)	(44.30)	(23.08)	(7.98)
Observations	1,084,052	1,084,114	1,313,073	1,182,226	1,181,733	971,897
R-squared	0.748	0.752	0.767	0.757	0.705	0.692
Controls	Yes	Yes	Yes	Yes	Yes	Yes
prefecture × year FE	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Distance	< 30km	< 30km	< 30km	< 30km	< 30km	< 30km

Note: I take log of all the outcome variables. Control variables include the age of the firm and also polynomial terms of the distance. The standard errors are clustered on 0.1° latitude × 0.1° longitude block, *** p<0.01, ** p<0.05, * p<0.1

Table 9: with different controls

	(1) sales	(2) employees	(3) production	(4) asset	(5) sales cost	(6) admin expenses	(7) profit
treat	0.053**	0.043**	0.054**	0.061***	0.052**	0.057**	0.016
	(2.21)	(1.98)	(2.24)	(3.31)	(2.10)	(2.01)	(0.36)
dist	0.014*	0.010	0.014*	0.010	0.014*	-0.010	0.003
	(1.83)	(1.62)	(1.83)	(1.45)	(1.92)	(-0.95)	(0.22)
treat × dist	0.002	-0.001	0.002	-0.001	0.001	-0.000	0.008
	(0.83)	(-0.37)	(0.84)	(-0.75)	(0.21)	(-0.17)	(1.61)
Constant	11.041***	5.507***	11.036***	10.599***	10.982***	7.824***	7.277***
	(35.74)	(20.39)	(35.72)	(48.91)	(24.90)	(25.18)	(17.66)
Observations	603,929	693,284	603,988	736,065	656,828	656,498	561,097
R-squared	0.827	0.820	0.831	0.875	0.830	0.806	0.751
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
prefecture × year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Distance	< 30km	< 30km	< 30km	< 30km	< 30km	< 30km	< 30km

Note: I take log of all the outcome variables. Control variables include the age of the firm and also polynomial terms of the distance. The standard errors are clustered on 0.1° latitude × 0.1° longitude block, *** p<0.01, ** p<0.05, * p<0.1